Kasi Yedumati · Full Stack Training

Lesson 4 — Response Models and Error Handling

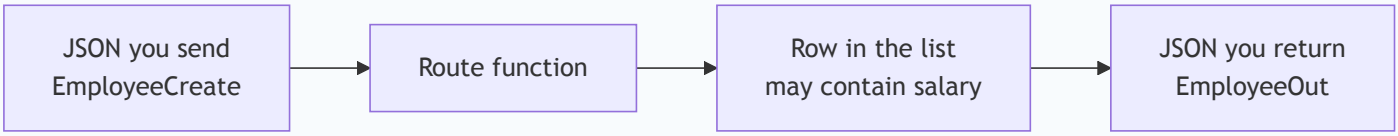
Course: FastAPI for Full Stack · **Session 4 of 10** · Learning path · **Previous:** Lesson 3 · **Next:** Lesson 5

> **Module Outcome:** You can return a directory card without extra fields, reject a bad partial update, and send 404 when the employee id is missing.

You will build	Separate shapes for the JSON you accept and the JSON you return
New words	Response model, input schema, output schema
You need	`EmployeeCreate` and the in-memory `employees` list in `main.py`

From the requirements. The directory card shows name, designation, and department. The profile adds email and phone. Salary is not on those screens. If a row in memory has a salary, the API must not send it.

1. Two shapes



`EmployeeCreate` is what a registration must contain. `EmployeeOut` is what a GET is allowed to contain. `response_model` builds the outgoing JSON from `EmployeeOut` and drops every other key.

Ananya's row in the list now includes `"salary": 90000`. That key stays in memory. It does not leave the API.

Add these two classes under `EmployeeCreate`. The whole file is `examples/lesson04/01_response_model.py`.

```
from pydantic import BaseModel, ConfigDict, EmailStr, Field, field_validator

class EmployeeUpdate(BaseModel):
    name: str | None = Field(default=None, min_length=1)
    designation: str | None = None
    department: str | None = None
    email: EmailStr | None = None
    phone: str | None = None

    @field_validator("name", "designation", "department", "email")
    @classmethod
    def not_null(cls, value):
        if value is None:
            raise ValueError("cannot be null")
        return value

class EmployeeOut(BaseModel):
    model_config = ConfigDict(from_attributes=True)

    employee_id: str
    name: str
```

```
    designation: str
    department: str
    email: EmailStr
    phone: str | None = None
```

Line	Job
`EmployeeUpdate`	A PATCH body. Every field is optional. Only the keys you send are changed
`not_null`	A key you leave out is fine. A key you send as `null` is **422**, except `phone`, which may be cleared
`EmployeeOut`	The fields a caller is allowed to see
`from_attributes=True`	Not needed yet: a dictionary works without it. It lets the model read a database row object in Lesson 5

Put `response_model=EmployeeOut` on the GET, POST, PUT, and PATCH routes. The list uses `list[EmployeeOut]`.

```
@app.get("/employees/{employee_id}", response_model=EmployeeOut)
def get_employee(employee_id: str):
    return find_employee(employee_id)
```

Open `GET /employees/E-1042`.

Output:

```
{
  "employee_id": "E-1042",
  "name": "Ananya Rao",
  "designation": "Senior Software Engineer",
  "department": "Engineering",
  "email": "ananya.rao@acme.example",
  "phone": "+91 90000 01042"
}
```

There is no `salary` key.

2. PATCH checks the fields, and 404 is still yours

Change `update_employee` so the body is `EmployeeUpdate`, not `dict`.

```
@app.patch("/employees/{employee_id}", response_model=EmployeeOut)
def update_employee(employee_id: str, changes: EmployeeUpdate):
    current = find_employee(employee_id)
    current.update(changes.model_dump(exclude_unset=True))
    current["employee_id"] = employee_id
    return current
```

`exclude_unset=True` leaves out the keys the caller did not send, so `update` only touches the keys that were sent. A PATCH of `{"designation": "Lead Engineer"}` does not wipe the email.

A PATCH with `"email": "not-an-email"` is **422**, and the row does not change. `find_employee` still raises **404** when the id is missing. That exception becomes:

```
{"detail":"Employee not found"}
```

Common error: PATCH uses `model_dump()` without `exclude_unset=True`. Missing keys are sent as `null` and the stored email becomes `null`.

Common error: `EmployeeUpdate` has no `not_null` check, and a PATCH sends `{"name": null}`. The row now holds `None`, and every GET of that employee, and the whole list, is **500**, because `EmployeeOut` says `name` is a string. Leaving a key out and sending `null` are different requests.

Common error: The GET route has no `response_model`, so `"salary": 90000` appears in the browser. The class `EmployeeOut` does nothing until a route names it.

Try these

1. **(Easy)** GET Ananya. Confirm `salary` is absent.
2. **(Easy)** PATCH her designation. GET again. Email and phone are unchanged.
3. **(Medium)** PATCH with a bad email. Status is 422. The designation from step 2 is still there.
4. **(Stretch)** POST a new employee. The **201** body is an `EmployeeOut`. It has no salary key even if you add one in the stored dict afterwards.

Interview: *Why two models?* — The caller may send a registration, but the directory is only allowed to show the public fields. `response_model` is that rule.

Interview: *Who sends 404?* — Your code, with `HTTPException`, when the id is not in the list. FastAPI sends 422 when the body fails the model.

Key takeaways

- `EmployeeCreate` checks input. `EmployeeOut` limits output. `EmployeeUpdate` is a partial change.
- `response_model` drops keys that are not on the output model.
- `exclude_unset=True` is what makes PATCH partial.
- 404 is `HTTPException`. 422 is a failed model.

Lab — Ticket ACME-204

The directory must not show salary, and a partial update must not accept a broken email.

```
PYTHON/17_FastAPI/labs/L04_response_model/
├── broken/main.py
├── starter/main.py
└── solution/main.py
```

Part A — Fix the broken API (10 min)

`broken/main.py` starts. It has **3 bugs**:

1. `GET /employees/E-1042` includes `"salary": 90000`.
2. `PATCH` with `"email": "not-an-email"` is stored.
3. `GET /employees/E-9999` returns `null` and **200**.

Part B — Build the models (15 min)

`starter/main.py` registers employees with `EmployeeCreate` and stores a dictionary. Add `EmployeeOut` and `EmployeeUpdate`, and set `response_model` on the read and write routes. Add `"salary": 90000` to Ananya's dictionary so a GET can prove that key stays inside the API.

Part C — Prove it works (5 min)

- [ ] GET Ananya has no `salary` key.
- [ ] PATCH designation only. Email stays `ananya.rao@acme.example`.
- [ ] PATCH `"email": "not-an-email"` returns **422**.
- [ ] PATCH `{"name": null}` returns **422**, and `GET /employees` is still **200**.
- [ ] GET `E-9999` returns **404** and `{"detail":"Employee not found"}`.

Open `solution/main.py` only after the checklist passes.

Homework

1. Finish the lab in `acmehr-api`.
2. **Transfer drill (15 minutes).** Add a `notes` key on one employee. GET that employee and confirm `notes` is not in the JSON. Remove the key afterwards if you do not want it in the list.

Next: Database connectivity. The list moves into MySQL. `EmployeeOut` stays.